

Energy Conversion I

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Desk D216

Layout

- Non isolated choppers
- Switch mode power supplies
- Power components
- **Sinusoidal absorption**



Introduction

Boost

Flyback

Example of converter



Introduction

Introduction

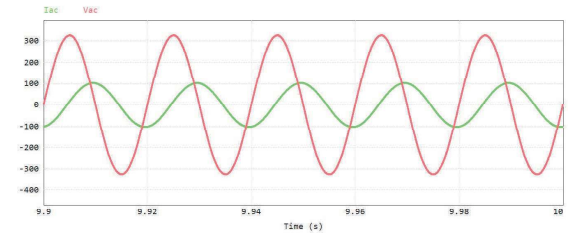
Boost

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Reminder: Power in AC

$$V_{RMS} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$



- **Real power:** $P = \langle p(t) \rangle = \langle v(t) \cdot i(t) \rangle = V_{RMS} \cdot I_{RMS} \cdot \cos(\varphi_1)$
- **Reactive power:** $Q = V_{RMS} \cdot I_{RMS} \cdot \sin(\varphi_1)$
- **Apparent power:** $S = V_{RMS} \cdot I_{RMS}$
- **Power factor:** $pf = \frac{P}{S}$
- **Total harmonic distortion:** $THD = \frac{\sqrt{\sum_{n=2}^{\infty} I_{n,RMS}^2}}{I_{1,RMS}}$

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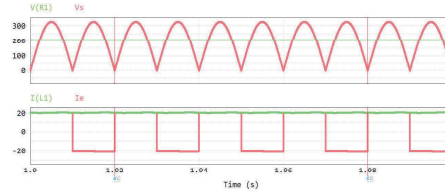
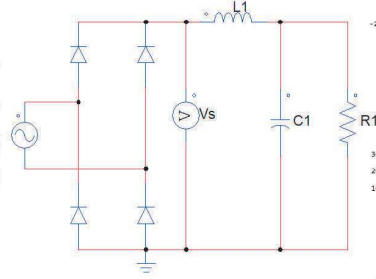
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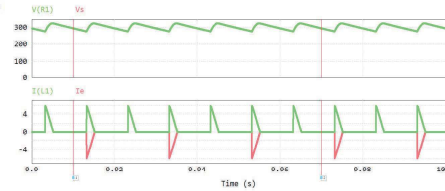
Example of converter



Reminder: Diode rectifier



$$L = 10H \rightarrow pf = 0,9 \quad THD = 0,48$$



$$L = 10nH \rightarrow pf = 0,5 \quad THD = 1,5$$

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European standard: 61000-3-2

Limits for harmonics currents emission (class C ≈ lightning)

Tableau 2 – Limites pour les appareils de classe C

Rang harmonique	Courant harmonique maximal exprimé en pourcentage du courant fondamental d'entrée des luminaires
n	%
2	2
3	30
5	10
7	7
9	5
11 à n ≤ 39 (harmoniques impairs seulement)	3

* C'est le facteur de puissance du circuit

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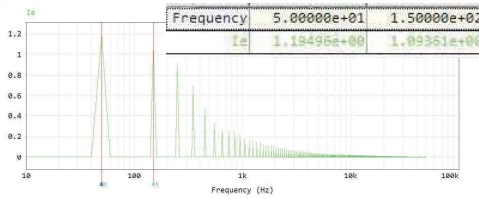
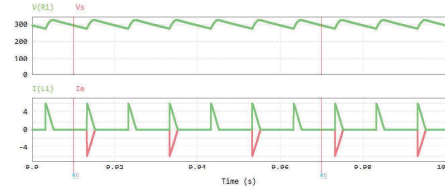
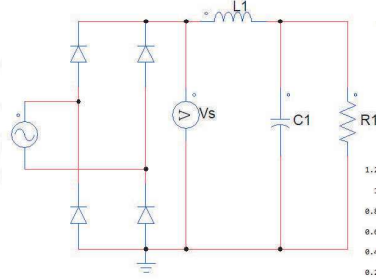
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European standard: 61000-3-2



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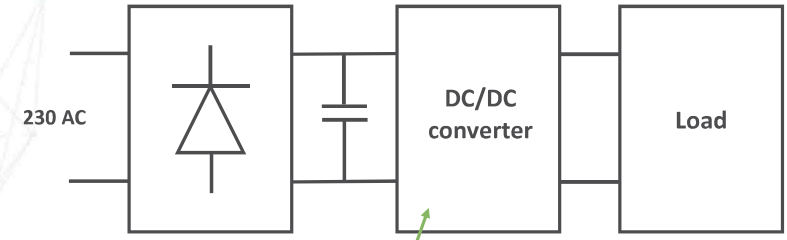
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Power factor correction

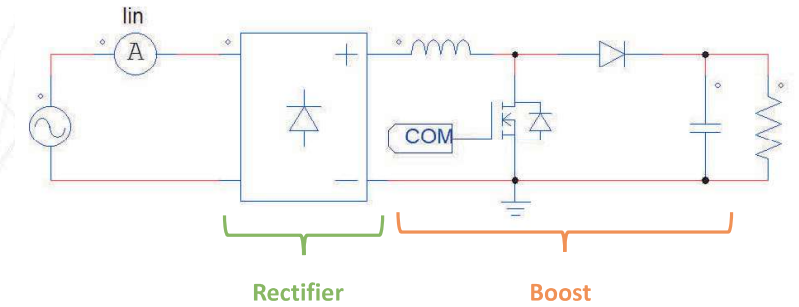


Used to absorb sinus current (before rectifier)

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Boost for PFC

Structure



Structure

$$v_r = V\sqrt{2} \sin(\theta)$$

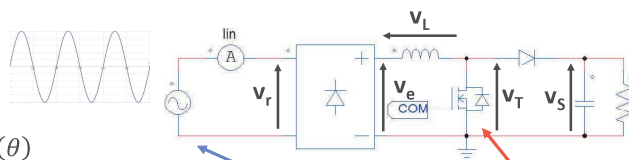
$$\theta = 2\pi f_{BF} t$$

Goal:

$$I_{in} = I_r \sqrt{2} \sin(\theta)$$

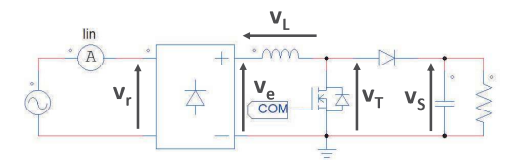
$$v_e = |v_r| \text{ and } v_e = (1 - \alpha) v_s \rightarrow V_s > v_{r \max}$$

- Two control strategies :
 - Fixed switching frequency -> PWM
 - Free switching frequency -> hysteresis control



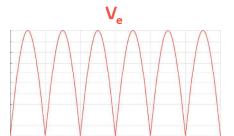
- 2 frequencies: f_{LF} (50Hz for grid) and f_{HF} (few 10 kHz for switching)

PWM control



- With PWM control, the switching frequency is fixed, the duty cycle is used to control the current
- For low frequency, the boost is a continuous voltage amplifier controlled with the duty cycle α

- Continuous conduction of diodes in the rectifier:
 $v_e = |v_r| = V\sqrt{2} \cdot |\sin(\theta)|$



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Boost

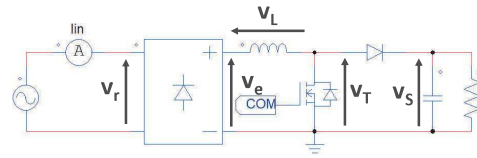
Structure
PWM control
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Example of converter



PWM control



- $v_L = v_e - v_T = V\sqrt{2} \cdot |\sin(\theta)| - (1 - \alpha)V_S = L \frac{di_L}{dt}$
- We want $i_{in} = I \cdot \sqrt{2} \sin(\theta)$
 - $i_L = I \cdot \sqrt{2} \cdot |\sin(\theta)|$
- For $0 < \theta < \pi$:
 - $1 - \alpha = \frac{V\sqrt{2}}{V_S} \cdot \sin(\theta) - \frac{L \cdot 2\pi f_{LF} \cdot I \sqrt{2}}{V_S} \cos(\theta)$
 - $\alpha = 1 - \frac{V\sqrt{2}}{V_S} \cdot \left(\sin(\theta) - \frac{2\pi L f_{LF} P}{V^2} \cos(\theta) \right)$

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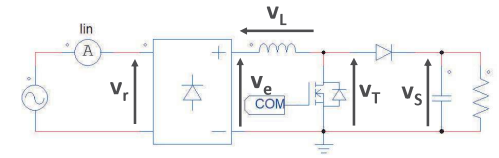
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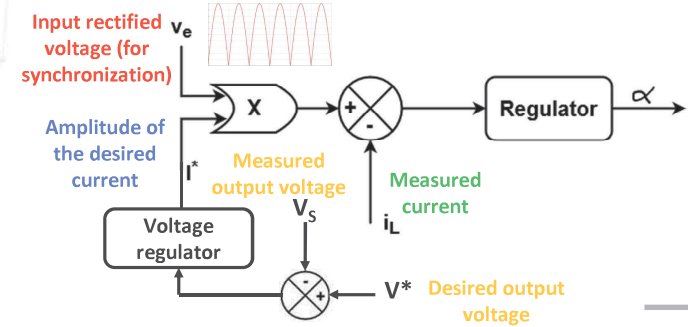
Example of converter



PWM control



$$\alpha = 1 - \frac{V\sqrt{2}}{V_S} \cdot \left(\sin(\theta) - \frac{2\pi L f_{LF} P}{V^2} \cos(\theta) \right)$$



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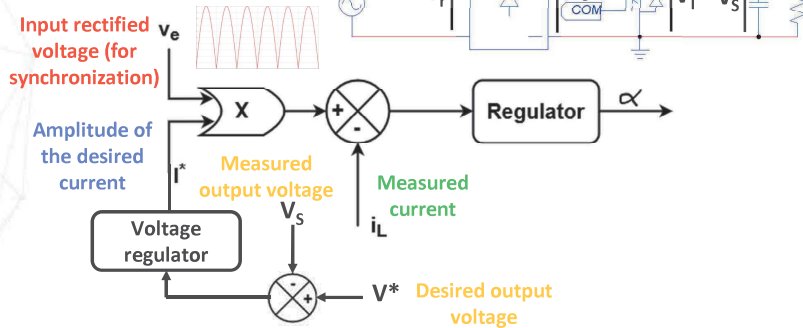
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PWM control



Condition: voltage control « slow » (few Hz) to ensure sinusoidal current

Filter the output voltage for measurement (sample at $2f_{BF}$ band-stop filter, high output capacitor)

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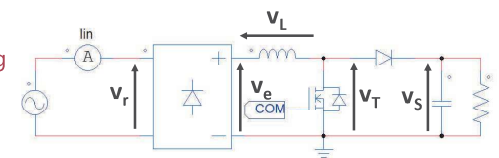
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PWM control: components sizing



- Semi-conductors

$$\langle i_D \rangle = \frac{P}{V_S} \quad \langle i_T \rangle = \langle i_L \rangle - \langle i_D \rangle = \frac{P}{V_S} \left(\frac{4}{\pi} \frac{V_S}{V\sqrt{2}} - 1 \right)$$

$$I_{T\text{ RMS}} = \frac{P}{V} \sqrt{1 - \frac{8 \cdot V\sqrt{2}}{3\pi S}} I_{D\text{ RMS}} = \frac{P}{V} \sqrt{\frac{8 \cdot V\sqrt{2}}{3\pi V_S}}$$

- Output capacitor

$$\Delta V_{S\text{ BF}} = \frac{P}{2\pi f_{BF} C V_S} \quad I_{C\text{ RMS}} = \frac{P}{V_S} \sqrt{\frac{16 \cdot V_S}{3\pi V\sqrt{2}}} - 1$$

- Input inductor: limit the current ripple due to switching BUT small voltage drop

$$\Delta i_{HF} = \frac{V_S}{4L f_{HF}} \quad \frac{V}{2\pi I f_{BF}} \gg L > \frac{V_S}{4f_{HF} \Delta i_{HF}} \quad I_{L\text{ RMS}} = \frac{P}{V}$$

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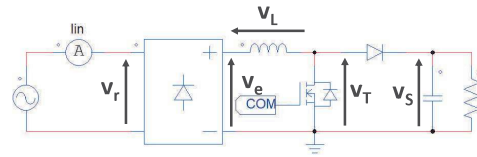
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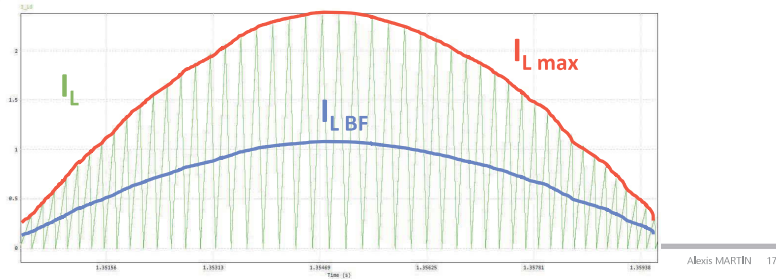


Hysteresis control



PWM control: need a current regulator to follow a sinus input
➤ Complicated to design the regulator

➤ Hysteresis control



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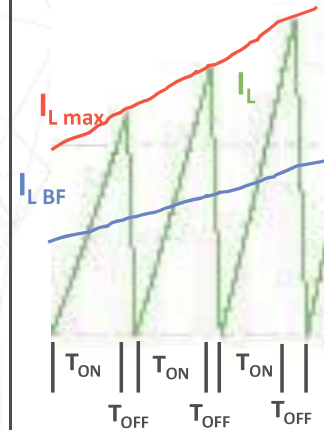
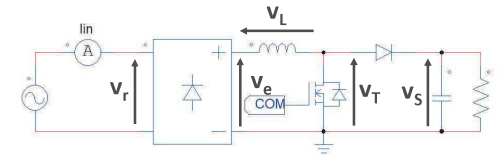
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Hysteresis control



- $I_{L \max} = \frac{V_e}{L} t_{ON} = 2I_{L \text{ BF}} = 2I\sqrt{2}|\sin(\theta)|$
- $t_{ON} = \frac{2LI}{V} = \frac{2LP}{V^2} = cst$
- $t_{OFF} = t_{ON} \frac{1-\alpha}{\alpha} = t_{ON} \frac{V_e}{V_S - V_e}$
- $f_{HF} = \frac{1}{t_{ON} + t_{OFF}} = \frac{1}{t_{ON}} \left(1 - \frac{V\sqrt{2}}{V_S} |\sin(\theta)|\right)$
- $f_{HF \min} = \frac{1}{t_{ON}}$
- $f_{HF \max} = \frac{1}{t_{ON}} \left(1 - \frac{V\sqrt{2}}{V_S}\right)$

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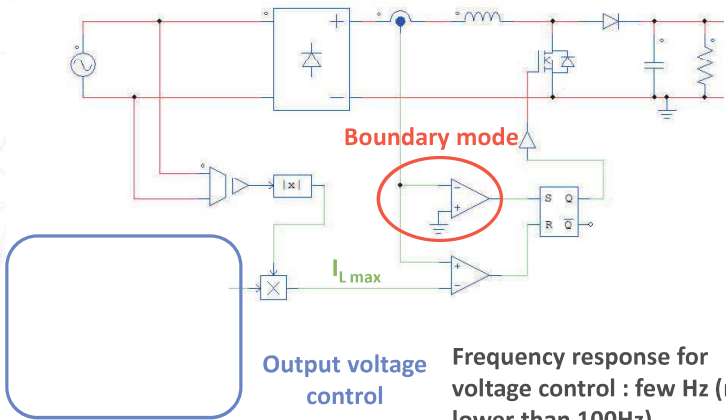
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Hysteresis control



Output voltage control

Frequency response for voltage control : few Hz (much lower than 100Hz)

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Example: LED driver

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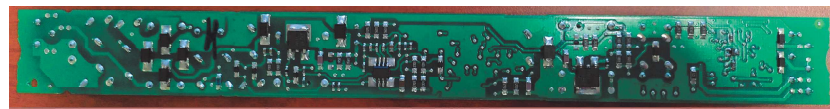
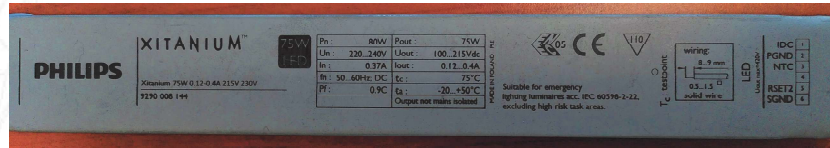
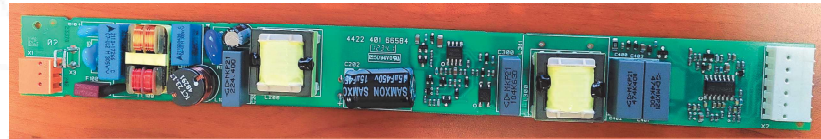
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Example: LED driver



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Flyback for PFC

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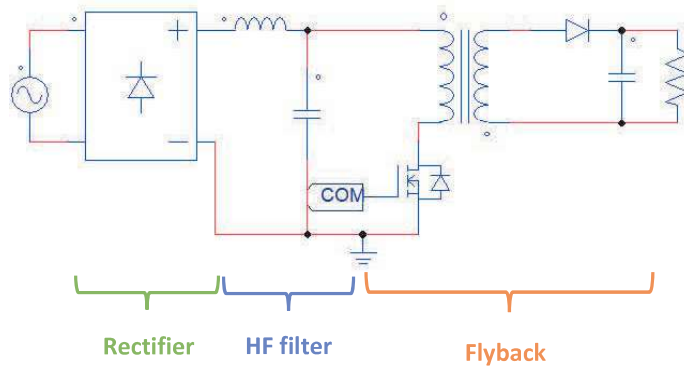
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Example of RT7306
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Structure



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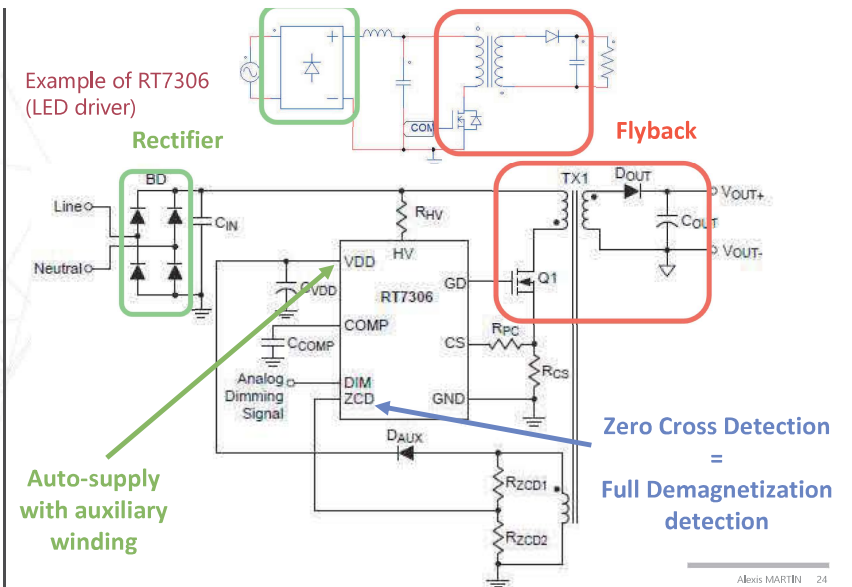
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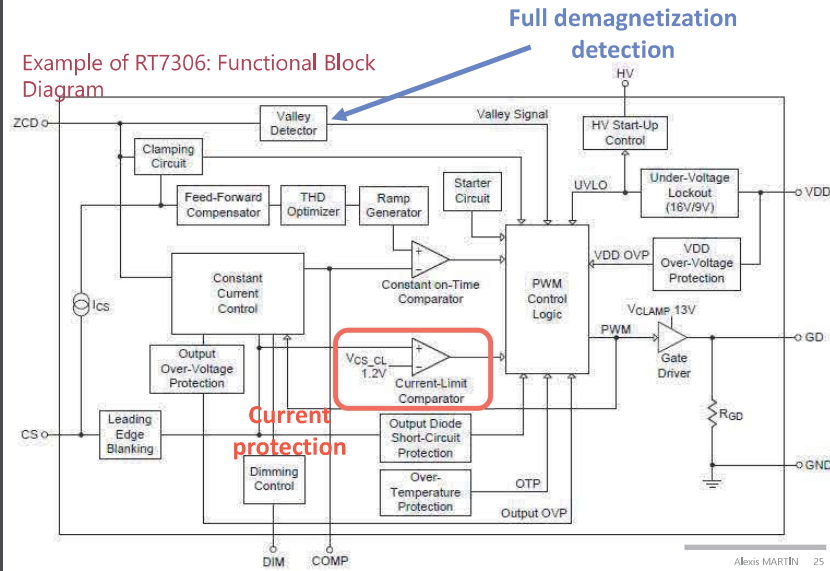
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Example of RT7306
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Example of RT7306: Functional Block
Diagram



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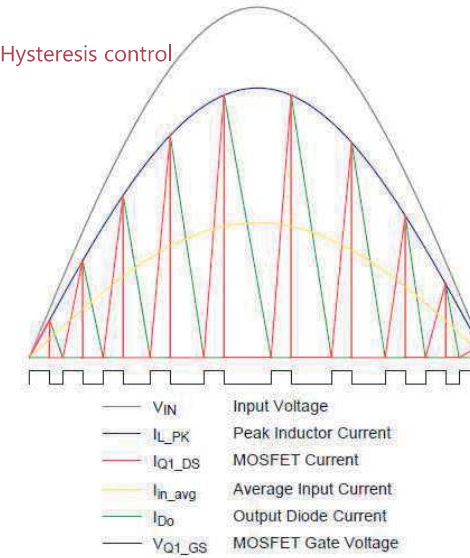
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Hysteresis control



$$I_{L_PK} = \frac{V_{IN}}{L_m} \times t_{ON}$$

$$I_{L_PK} = \frac{V_{IN_PK} \times |\sin(\theta)| \times t_{ON}}{L_m}$$

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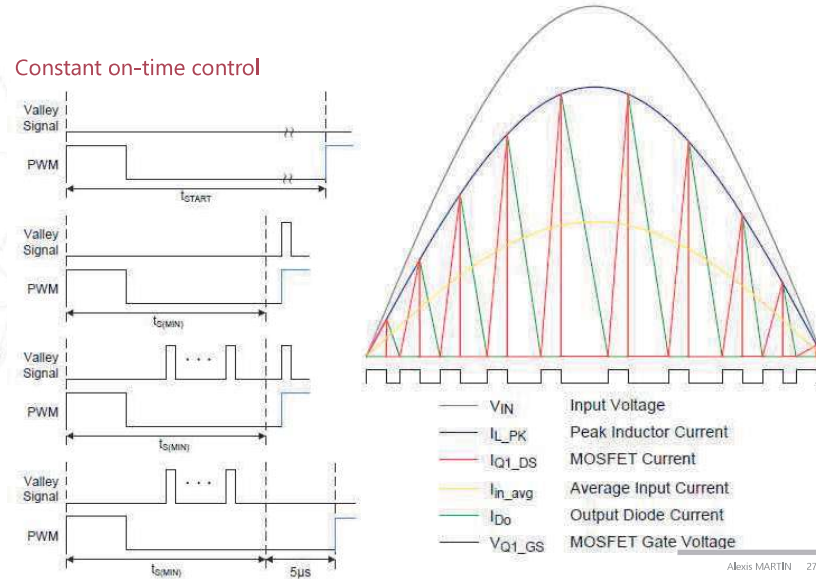
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Constant on-time control



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Example of a full converter

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Example: EVL400W-80PL



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Example: EVL400W-80PL

The architecture is based on a two-stage approach: a front-end PFC pre-regulator based on a CCM (Continuous Conduction Mode) boost PFC controller using the L4985A, and a downstream LLC resonant half-bridge converter, designed around the L6699.

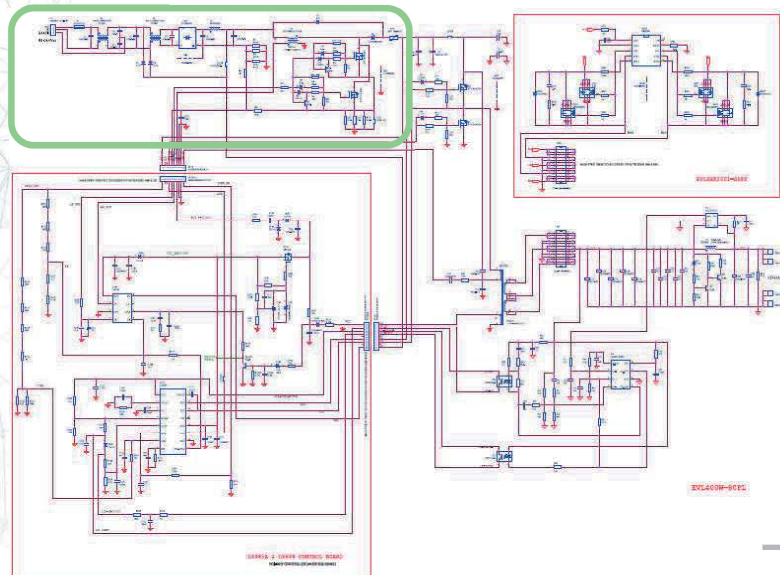
Universal input mains voltage range: from 90 Vac to 264 Vac – frequency from 45 to 65 Hz

Output voltage: 12 V at 33 A continuous operation

- Overall efficiency at full load: > 89%, according to ENERGY STAR® 6.1 limit for computer and compliant with 80Plus PLATINUM level
- Average efficiency: > 89%, according to European CoC ver. 5 Tier 2 for external power supplies
- Efficiency at 250 mW > 50%, compliant to EuP lot 6 Tier 2 limit for household and office equipment
- No load mains consumption: < 150 mW at 230 Vac, below European CoC ver. 5 Tier 2 limit for external power supplies
- Mains harmonics: meets EN-61000-3-2 Class-D and JEITA-MITI Class-D
- EMI: according to EN55022 Class-B
- Safety: meets EN60950 standards
- RoHS compliant

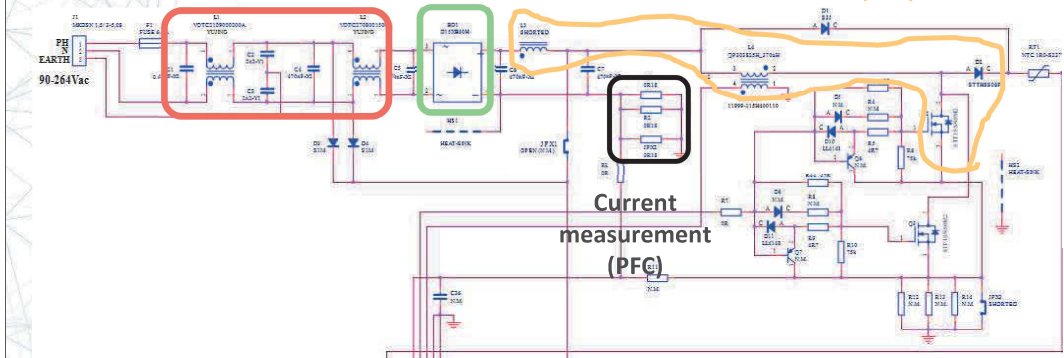


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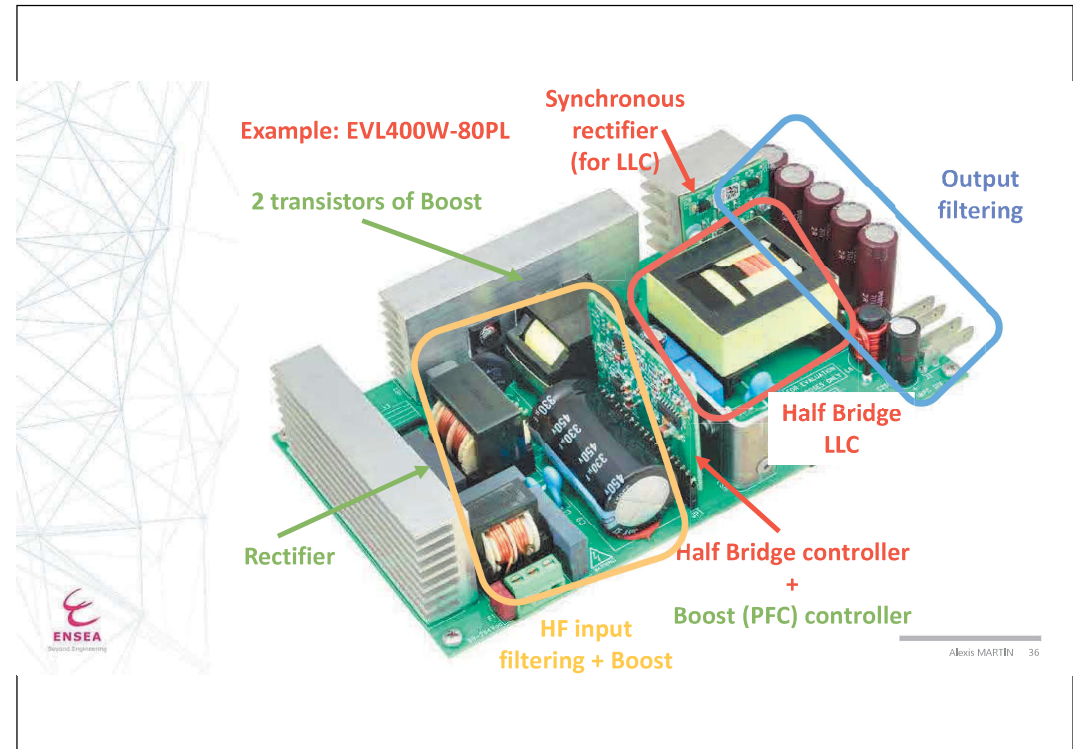
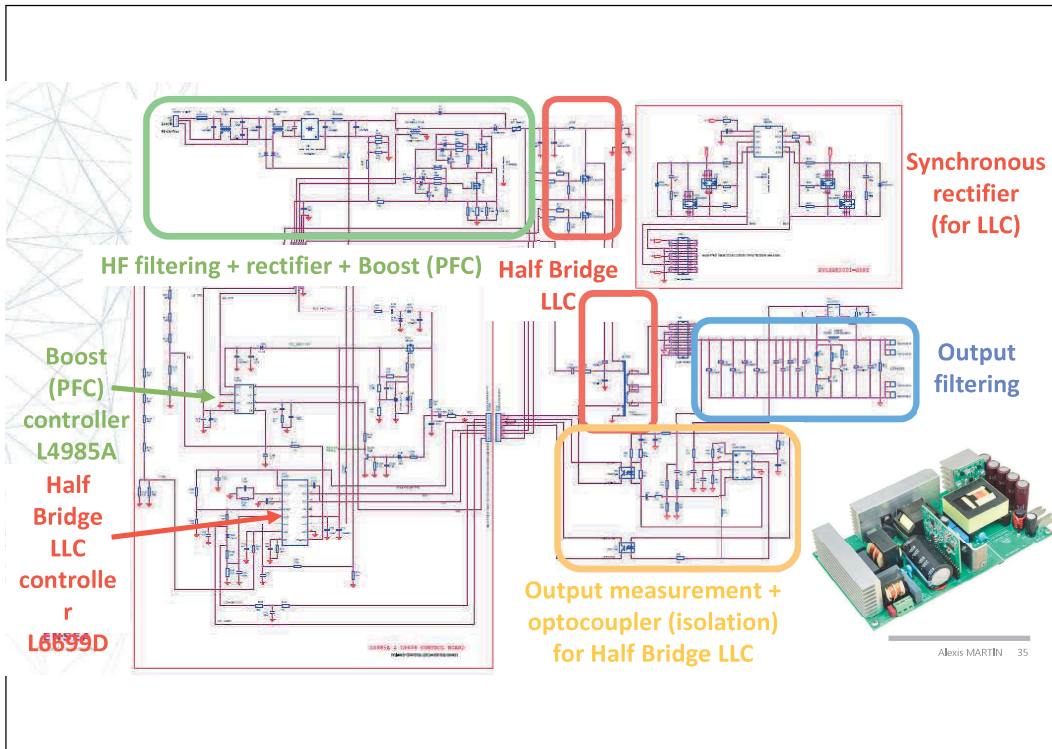
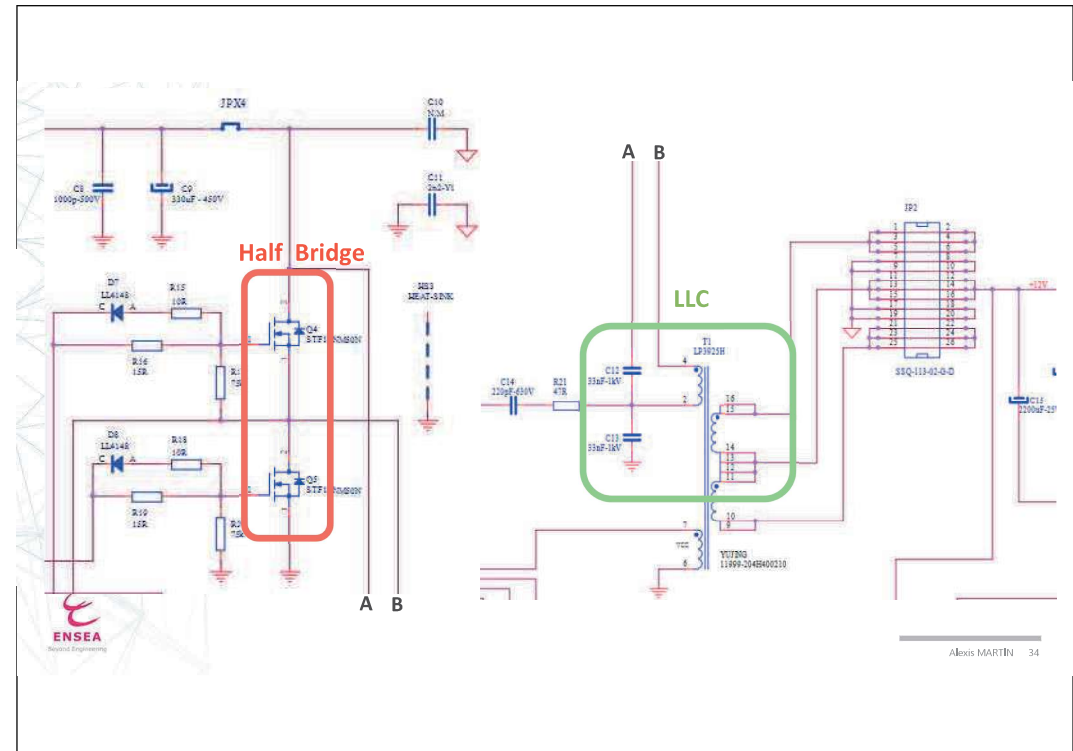
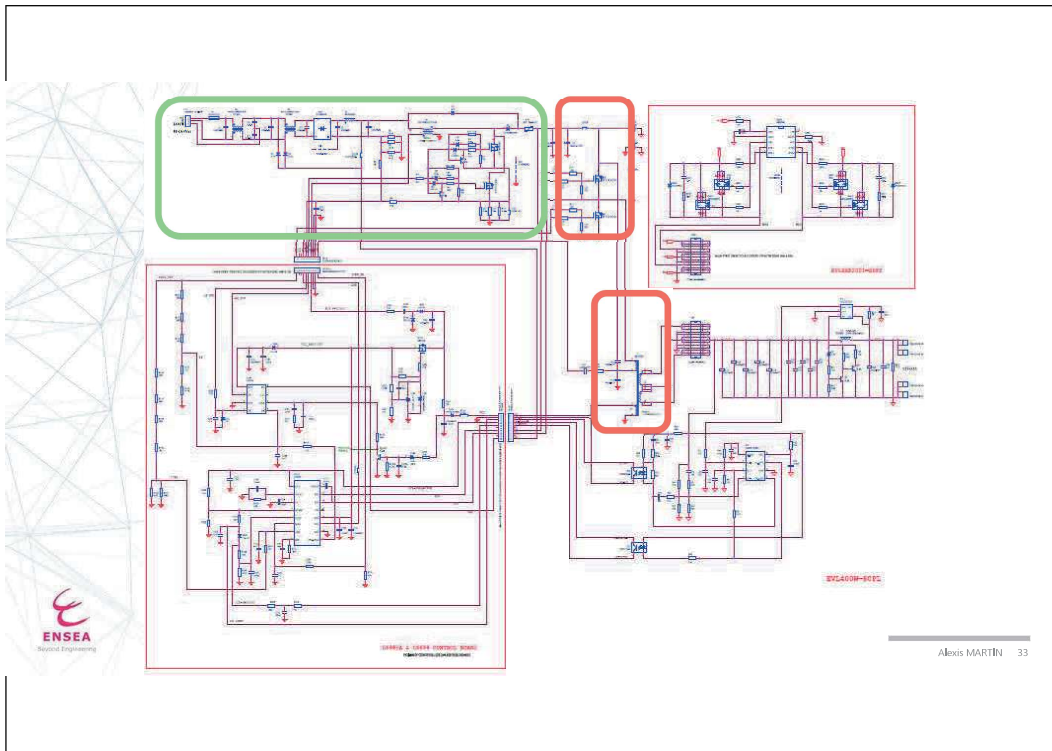


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HF filtering Rectifier Boost (PFC)



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Example: EVL400W-80PL

